Carbon Neutrality

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# Multi-period superstructure optimization for CCS source-sink matching in South Sumatra industrial clusters

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**Abstract**

Minimizing supply chain costs is a critical challenge for the successful deployment of full-chain $CO_2$ capture and storage (CCS) systems, particularly in industrial clusters with diverse source and sink characteristics. This study presents a superstructure optimization model using Mixed-Integer Linear Programming (MILP), focusing on a case study in South Sumatra, Indonesia. The model's key innovation is the integration of time-dependent reservoir injectivity profiles, derived from well simulations and decline analysis, directly into the MILP framework. This allows the model to constrain injection timing and capacity, accurately reflecting reservoir availability over a 30-year planning horizon. The model evaluates multiple network configurations, including single-well and multi-well systems, allowing for coordinated injection scheduling. The results highlight the critical role of Hub-based infrastructure, where optimized configurations reduce transportation costs by approximately 34%, from an average of $25.36 to $16.72 per ton, through economies of scale and shared infrastructure. Multi-well systems further improve storage capacity and injection efficiency, particularly when injection schedules are carefully optimized. By strategically managing injection capacity and scheduling reservoir availability, the system achieves additional cost reductions, with transportation costs ranging from $15.89 to $16.56 per ton. The findings highlight the critical role of superstructure-based designs in optimizing the interactions among sources, sinks, and transportation networks. Coordinated injection strategies and infrastructure planning are essential to reducing CCS costs and enhancing scalability. This study underscores the value of systematic optimization in achieving cost-effective and scalable carbon storage solutions across diverse industrial contexts.

**Keywords** CCS, CCS hub, MILP, Superstructure, Optimization, Industrial cluster

## 1 Introduction

Global warming has become a major concern, primarily driven by the increasing greenhouse gas (GHG) emissions over recent decades. Emissions from fossil fuels are recognized as key contributors to global warming and significant changes in climate patterns [20]. Among anthropogenic GHGs, carbon dioxide ($CO_2$) is the primary contributor, with global emissions reaching a historic high of 37.4 Gt in 2023 [21]. As a response, Carbon capture and storage (CCS) offers a viable solution to mitigate $CO_2$ emissions from major sources, such as power plants and cement manufacturing facilities. This process involves capturing $CO_2$, transporting it through

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dedicated pipelines, and sequestering it underground [6]. By 2023, global CCS infrastructure had reached an annual storage capacity of 64.8 million tons across 41 projects [17].

The selection of CCS as the focus of this study reflects its critical and irreplaceable role in climate mitigation, especially for hard-to-abate sectors. Recent studies highlight that without industrial CCS, deep decarbonization in key sectors is extremely difficult, and the cost of climate policy increases substantially. CCS thus becomes a critical mitigation option to support both continued industrial production and achievement of net zero goals [36]. Large-scale studies conducted in China further support this view, highlighting that CCS is the only practical technological solution for achieving substantial emission reductions from coal-fired power plants when deployment is undertaken at an appropriate stage, supported by strong policy incentives and optimized source–sink integration [11, 13, 30]. In Indonesia, concentrated emission sources and extensive geological storage capacity underscores the significance of CCS. Effective system design and timely deployment of CCS are therefore pivotal to supporting the country's net zero pathway and ensuring continued industrial growth alongside substantial emissions reductions.

Despite its strategic importance, large-scale CCS deployment faces significant challenges in emerging economies like Indonesia. Integrating dispersed emission sources with suitable storage sites requires complex infrastructure and careful network planning within CCS clusters. Key barriers include the identification and characterization of viable storage sites, the development of extensive pipeline systems, and the management of high capital and operational costs across the CCS value chain [12, 28]. Addressing the source-sink matching problem, which involves connecting captured $CO_2$ to suitable storage sites for sequestration, is therefore crucial. Optimizing the connections between $CO_2$ sources and sinks can lead to more efficient CCS solutions.

Recent advancements in large-scale CCS systems have introduced various optimization methods. Middleton and Bielicki [31] developed a Mixed-Integer Linear Programming (MILP) model, SimCCS, that minimizes overall CCS costs and demonstrates the cost-effectiveness of network systems due to economies of scale. Tan et al. [50] created a continuous-time MILP optimization model for source-sink matching, but it did not account for reservoir injection limits––a critical aspect for practical applications. Subsequent models introduced discrete-time frameworks to address these injection limits [51]. In the Norwegian CCS chain, a multimode MILP approach incorporates multiple constraints, including mass balance and intermediate storage sizing [14]. Mixed-integer nonlinear programming (MINLP) has also been applied to optimize $CO_2$ pipeline networks in South Korea, revealing that strategically clustering emissions locally can be more cost-effective than long-distance transport [29]. Additionally, pinch-based CCS source-sink matching methods have been employed, highlighting the storage capacity and injection rate constraints [9, 32].

A promising advancement in CCS optimization is the use of superstructure-based approaches, which enable the simultaneous evaluation of all potential infrastructure and routing alternatives in one unified model, thus supporting more robust and systematic system optimization. For example, a case study in Northeastern China [56] applied a superstructure-based MILP model to optimize the integration of $CO_2$ capture units, transport pipelines, and geological storage sites, enabling simultaneous evaluation of various network configurations. Similarly, in Indonesia, superstructure optimization using tools such as a General Algebraic Modeling System (GAMS) with a nonlinear programming approach has been applied to identify optimal and cost-effective CCS/CCUS designs [10].

CCS supply chain models offer robust frameworks for defining objective functions, decision variables, and constraints. However, significant gaps persist in aligning these models with the dynamic behavior of reservoirs, particularly concerning injection rate limits and the timing of $CO_2$ injection. These elements are pivotal for maximizing storage capacity and influencing economic considerations in the optimization of injection operations [16].

One critical gap in the existing literature is the inadequate integration of technical reservoir behavior into the timing of $CO_2$ injection. For instance, both Norwegian CCS projects and a CCS initiative in China adopted predefined deployment intervals with 5-year system cycle. However, these studies did not provide a clear technical or operational justification of these assumptions [14, 30]. A South Korean case applied policy-based intervals to determine injection schedules but lacked a detailed explanation for their basis [29]. Some studies considered injection timing based on surface equipment readiness [52] yet failed to account for subsurface reservoir conditions. In practice, active reservoirs require complete depletion before being suitable for $CO_2$ injection [4]. Thus, injection planning must consider reservoir readiness to avoid operational conflicts and ensure storage integrity.

Beyond timing, the annual injection rate also plays a critical role in storage performance. A reservoir simulation study highlighted that the management of $CO_2$ injection rates can affect storage efficiency [4]. The storage efficiency is defined as the amount of $CO_2$ injected into

the reservoir relative to its total capacity. However, this study did not provide a comprehensive method for calculating the annual injection rates, which are essential for assessing long-term project viability. Some case studies simply divide total capacity by project duration [39, 56]. Additionally, some CCS studies have explored the effects of varying injection rates using an injectivity formula [3, 30]. These approaches fail to capture the dynamic nature of reservoirs, such as pressure decline and changing injectivity. This gap calls for models that integrate reservoir simulation outputs into planning tools.

To address those challenges, this study introduces a superstructure-based optimization framework for designing efficient CCS networks, with a focus on industrial clusters. In a superstructure model, all feasible facility connections and routing pathways are embedded into a unified network representation, enabling the optimization process to identify the most cost-effective configuration under spatial, temporal, and operational constraints. While superstructure-based optimization models have been widely used in previous CCS planning studies [56, 58], our approach further enhances this framework by integrating time-dependent reservoir injectivity and dynamic availability derived from well-level simulations. This enables annual injection scheduling in the optimization model to be directly informed by simulated reservoir performance, allowing for a more realistic and adaptable assessment of network operations.

The framework integrates capture units, pipelines, and geological storage sites into a MILP model, simulating $CO_2$ flow over a 30-year planning horizon. While existing studies often overlook critical interdependencies among injection rates, reservoir behavior, and activation timing, many are based on simplified assumptions such as static timing and proportional capacity allocation. This research makes several key contributions:

a) It incorporates injection scheduling based on reservoir depletion analysis, using simulated injectivity profiles to reflect the dynamic interplay between injection timing and reservoir performance; and
b) It refines capacity planning by employing annual injection calculations derived from well-level simulations, rather than static assumptions, enabling a more precise and reliable evaluation of storage efficiency.

Furthermore, three scenarios—single well, multiple wells, and delayed reservoir activation—are examined to evaluate their influence on network scalability and cost-effectiveness. By combining these advancements with a superstructure-based approach, this study offers a robust and technically grounded framework for CCS network optimization, contributing to the development of cost-effective and scalable carbon storage solutions.

## 2 Case study and superstructure model

### 2.1 CCS hub

This study examines South Sumatra due to its diverse industrial base, significant oil and gas reserves, and strategic suitability for $CO_2$ storage. The region hosts key industries, including power plants, cement factories, oil refineries, ammonia plants, and natural gas processing facilities. South Sumatra ranks as one of Indonesia's most promising $CO_2$ storage sites, benefiting from its stable geological structure and well-developed infrastructure [25]. The potential for CCS in this region has been further enhanced by growing international interest in cross-border CCS partnerships, particularly with countries such as Singapore and Japan [18].

$CO_2$ emissions from five industrial sectors, each with unique flue gas compositions, were analyzed in this study. Emissions from power plants, cement factories, and refineries result mainly from fuel combustion, whereas those from ammonia plants and natural gas processing facilities are linked to Acid Gas Removal Units (AGRU). Emissions from coal-fired power plants, refineries, and cement plants were calculated using stoichiometric methods, while combustion simulations were applied to gas-fired plants. $CO_2$ emissions from natural gas processing were sourced from [47] and emissions from ammonia plants were estimated analogously based on production capacity, as outlined in [46].

The planning of the CCS network involved calculating the annual $CO_2$ emissions from industrial sources, incorporating factors such as emission volumes, flue gas compositions, and geographical distributions. These calculations were instrumental in determining the scale of capture facilities, pipeline capacities, and the selection of optimal storage sites. The captured $CO_2$ undergoes dehydration to mitigate hydrate formation and pipeline corrosion, after which it is compressed to maintain a liquid or supercritical phase for efficient transport [34, 53]. $CO_2$ is then transported to geological storage sites, with a preference for depleted oil and gas reservoirs due to their proven integrity and suitability for long-term sequestration [35, 40].

Mapping the locations of industrial emission sources and storage reservoirs in South Sumatra revealed a close spatial relationship between the sources and suitable $CO_2$ sinks. As depicted in Fig. 1, the region comprises 14 emission sources generating 5.33 MTPA of $CO_2$, with the majority located in the eastern part, accounting for 55% of total emissions. The storage reservoirs, located primarily in the northern and southwestern regions, hold 78% of the total storage capacity. However, the readiness of

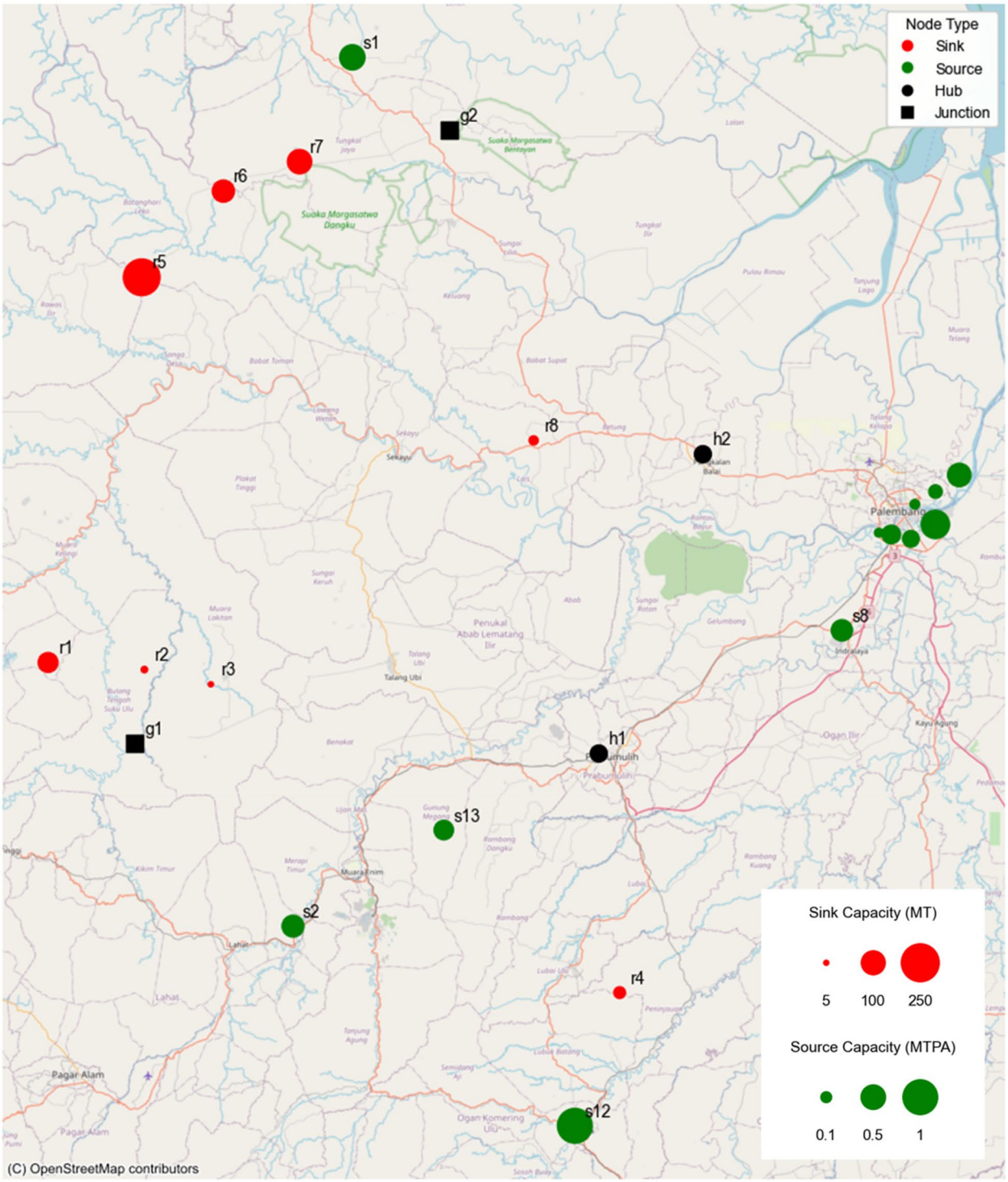



**Fig. 1** $CO_2$ emission sources and potential sinks in South Sumatra

these storage sinks varies due to factors such as current reservoir pressures and the performance characteristics of each reservoir. In this case, all oil and gas fields are still in production, which influences the availability and performance of reservoirs for $CO_2$ injection. This spatial distribution, combined with the operational dynamics of the reservoirs, emphasizes the importance of optimizing pipeline routing and strategically selecting storage sites

to minimize transportation distances and maximize storage capacity.

### 2.2 Superstructure representation

A superstructure representation refers to a generalized network framework that incorporates all feasible infrastructure elements and routing pathways into a unified model structure. This approach enables the optimization process to systematically evaluate and select among alternative source-sink connections, routing options, and facility activation strategies over time. Superstructure-based models have been widely adopted in previous CCS network optimization studies [56, 58], particularly for addressing complex routing decisions and infrastructure selection in large-scale CCS networks.

The superstructure of the CCS supply chain network is illustrated in Fig. 2, where all facilities are represented by a set of nodes, including emission sources (s), hubs serving as collection points (h), junction nodes (g) as endpoints of the hubs, and geological storage sites (r). Specifically, the system comprised 14 emission sources, two hubs, two junction nodes, and eight reservoir sinks. These elements were strategically mapped to form the foundation of a CCS network and illustrated as shown on Fig. 2.

The number and placement of hubs and junction nodes were guided by spatial and operational considerations. The two hubs represent strategic locations that currently serve as gathering points for oil and gas operations, with access to established infrastructure such as electrical supply, road networks, and pipeline right-of-way. Although not originally designed for $CO_2$ handling, these sites provide a practical foundation for integrating CCS infrastructure due to their existing logistical and utility advantages. Junction nodes were placed along existing pipeline corridors to improve access to dispersed storage sites and enable a modular network design that can accommodate future expansion.

In the proposed network configuration, Hub-11 (h1) and Hub-2 (h2) function as primary collection points for multiple pipelines, requiring larger dimensions to distribute $CO_2$ to the two junction nodes. From these junction nodes, smaller pipelines branch out toward the reservoir sinks. If the reservoir is more accessible from a hub than from a junction node, $CO_2$ may be routed directly from the hub to the reservoir. This flexible routing allows a single hub to serve both junction nodes and reservoirs, optimizing transportation routes and reducing overall system costs. The geographical distribution and storage capacities of the reservoirs are critical in determining

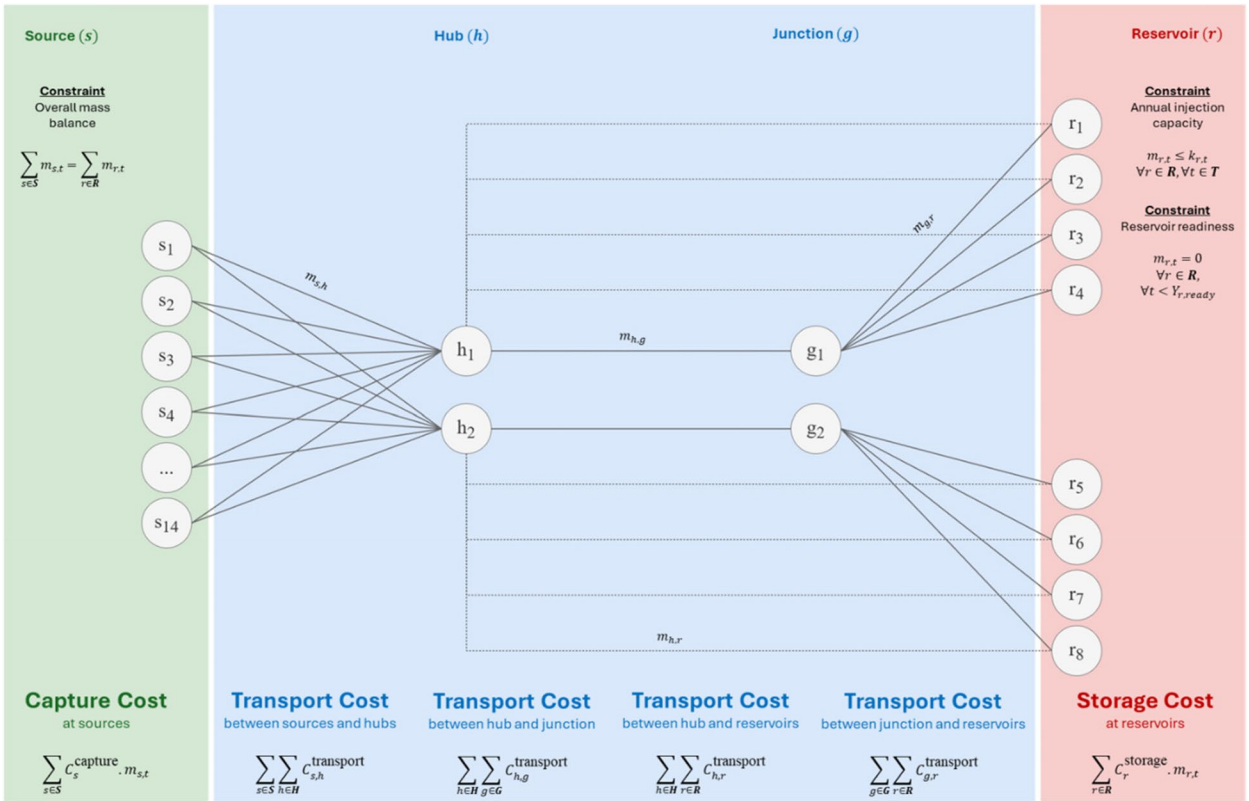



**Fig. 2** Superstructure framework

the optimal routing of $CO_2$, ensuring both efficient storage and long-term sustainability. Existing pipeline routes serve as a reference for developing new pipelines, leveraging pre-existing infrastructure and environmental assessments.

In this system, the availability of sinks may differ from that of sources, and each reservoir has an annual injection capacity that is influenced by its behavior. Based on this superstructure representation, a MILP problem was formulated to simulate $CO_2$ mass flows from sources to hubs, junction nodes, and sinks, as well as to evaluate the associated investment costs. The optimization model was designed for a planning horizon of 30 years.

## 3 Methodology

The research flow diagram (Fig. 3) depicts the sequential integration of $CO_2$ capture, transport, and storage modules within the CCS system. In this framework, each module comprises process simulation for capture, pipeline network modeling for transport, and reservoir

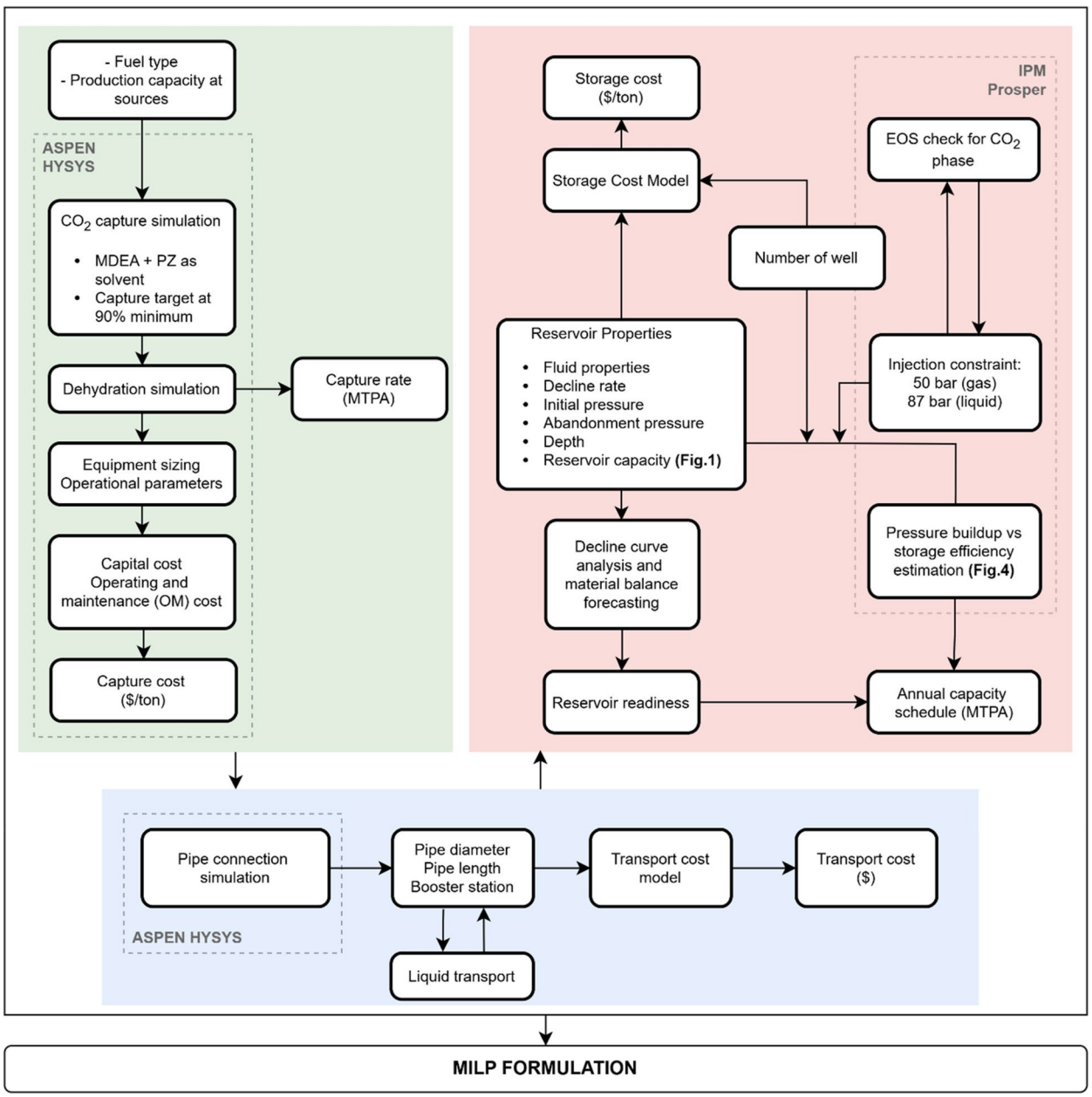



**Fig. 3** Research flow diagram for integrated optimization of the CCS system

analysis for injection and storage, all of which generate technical and economic parameters that are subsequently used as data input for the MILP optimization model. This approach ensures that the network configuration is optimized based on a consistent and coherent dataset across all components of the CCS system. The methodology for each module is described in detail in the following sections.

### 3.1 $CO_2$ emission sources

This study evaluated 14 emission sources representing diverse industrial sectors in South Sumatra. Process simulations for $CO_2$ capture were performed using ASPEN HYSYS V11. Chemical absorption using a mixture of methyl diethanolamine (MDEA) and piperazine (PZ) was selected to optimize $CO_2$ capture due to its high absorption capacity and rate, with PZ acting as a promoter for MDEA [54, 57]. The target capture rate was set at a minimum of 90% $CO_2$, and the operating parameters were obtained from a previous study [19]. The Acid Gas – Chemical Solvents property package in HYSYS was selected to accurately represent reaction kinetics.

For the dehydration process, the Glycol Package in ASPEN HYSYS V11 was used to simulate triethylene glycol (TEG) dehydration, with a glycol concentration set at 98.8% TEG by weight. Emission streams from AGRU units and ammonia plants, which already had high $CO_2$ content, were subjected only to dehydration without additional capture. Following capture and dehydration, the $CO_2$ stream was compressed to achieve conditions suitable for liquid or supercritical phase pipeline transport.

Outputs from the ASPEN HYSYS V11 simulations, such as compression power requirements, solvent circulation rates, and heat duties, were used to estimate equipment costs and operating expenses based on standard industry cost correlations. These costs were then levelized and expressed in USD per ton $CO_2$, and subsequently incorporated as fixed capture cost coefficients for each emission source in the MILP network optimization model.

### 3.2 Pipeline network

Captured $CO_2$ was transported to geological storage sites via a network of pipelines. Liquid or supercritical $CO_2$ was selected for transport to maximize efficiency. Pipeline operating temperatures were assumed to range from 13 to 44 degrees Celsius, and pressures between 85 and 150 bar [27]. The CCS network design included hubs as central collection points to channel $CO_2$ from multiple emission sources through large-diameter pipelines. From these hubs, $CO_2$ was routed either directly to storage reservoirs or via junction nodes, as determined by spatial optimization.

Pipeline network simulations were performed using ASPEN HYSYS V11 with the Peng Robinson Equation of State (EoS), which is appropriate for thermodynamic modeling of $CO_2$. Input parameters included $CO_2$ purity, inlet pressure and temperature, pipeline segment length, and pipe roughness. Simulation outputs, such as outlet pressure and phase state, were used to iteratively determine optimal pipeline diameters for each segment. Pipeline diameters for single-source connections ranged from 4 to 10 inches, and hub-to-node connections required larger diameters of 14 to 22 inches.

The simulation results provided the basis for estimating pipeline transportation costs, including both capital expenditures and operational expenditures. The distances and spatial relationships among all network nodes, including emission sources, hubs, junctions, and storage reservoirs, are provided in Supplementary Materials (Table S3).

### 3.3 MILP formulation

The proposed model aims to minimize the total annual cost (TAC) of the CCS supply chain network over a 30-year period. The CCS system consisted of 14 sources and eight reservoir sinks. The problem is formulated as a MILP using Python, solved with PuLP CBC. The TAC encompasses costs associated with $CO_2$ capture, transportation, and storage.

#### *3.3.1 Objective function*

$$\min(TAC) = \min \sum_{t \in T} \left[ \sum_{s \in S} C_s^{\text{capture}} . m_{s,t} + \sum_{s \in S} \sum_{h \in H} C_{s,h}^{\text{transport}} + \sum_{h \in H} \sum_{g \in G} C_{h,g}^{\text{transport}} + \sum_{g \in G} \sum_{r \in R} C_{g,r}^{\text{transport}} + \sum_{h \in H} \sum_{r \in R} C_{h,r}^{\text{transport}} + \sum_{r \in R} C_r^{\text{storage}} . m_{r,t} \right]$$

The total capture cost encompasses the expenses of $CO_2$ capture, dehydration, and compression from each source. It is calculated by multiplying the capture cost per ton ($C_s^{capture}$) by the annual $CO_2$ mass captured ($m_{s,t}$). The cost per ton ($C_s^{capture}$) was determined using technoeconomic analysis with investment costs based on the Guthrie modular method [45]. The ASPEN economic evaluations were used as inputs to calculate the bare module costs, which were adjusted to 2023 USD values.

The fixed and variable operation maintenance costs are assumed to be 3% each of the total capital investment [24]. The levelized cost method, assuming an 8% interest rate and 30-year plant lifespan, was used to express capture cost ($C_s^{capture}$) for each source in $ per ton. In this study, the estimated total capture cost ranges from $15 to $140 per ton of $CO_2$, depending on flue gas composition and plant scale. These values are consistent with the cost ranges found in recent CCS literature [28]. The capture cost for each source is provided in the Supplementary Materials (Table S1).

Economic evaluations of $CO_2$ pipelines often use different cost models, including linear, quadratic, and mass flow rate approaches, as reviewed in previous studies [7, 15]. In this study, transportation cost ($C^{transport}$) was based on the IEA-GHG cost model adjusted to 2023 USD [22]. This model applies a quadratic relationship to estimate investment costs based on pipeline diameter ($d$) and length ($l$). To integrate these costs into the MILP optimization framework, the investment costs for both pipelines and booster are calculated based on variations in ($d$) and ($l$) from pipeline simulations. Annual transportation costs ($C^{transport}$) were then computed using the transportation investment annuity method with an 8% interest rate and operational lifespan aligned with reservoir availability. The transportation costs are divided into the following sections: from source $s$ to hub $h$ ($C_{s,h}^{transport}$), from hub $h$ to junction $g$ ($C_{h,g}^{transport}$), from junction $g$ to reservoir $r$ ($C_{g,r}^{transport}$), and from hub $h$ directly to reservoir $r$ ($C_{h,r}^{transport}$) if no junction nodes are used. The total transportation cost is the sum of the annualized capital investment and yearly operational expenses for each section.

The cost of storage injection primarily includes the expenses for injecting supercritical $CO_2$ underground, drilling new wells, and conducting site monitoring [56]. Storage costs can vary significantly due to the heterogeneous properties of reservoirs [42]. In onshore depleted reservoirs, storage costs range from 1 to 10 EUR (2009) per ton of $CO_2$, depending on whether existing production wells are converted into injection wells or new wells are drilled [23]. This study applied the model developed by IEAGHG [22], with values adjusted to 2023 USD based on an 8% interest rate and operational lifespan determined by reservoir availability. The resulting storage cost estimates for this study range from $2 to $11 per ton of $CO_2$, which is consistent with the values reported in previous CCS studies. The total storage cost was calculated by multiplying the storage cost per ton ($C_r^{storage}$) by the $CO_2$ volume injected into each reservoir $r$ in tons ($m_{r,t}$).

#### *3.3.2 Constraints*

The captured $CO_2$ was transported and permanently stored in depleted oil and gas fields. In CCS planning, key factors include aligning $CO_2$ sources with storage sites based on time availability, $CO_2$ storage capacity, and injection rate [50]. $CO_2$ storage capacity was estimated in previous studies [55]. Determining the optimal injection start time is crucial and requires forecasting production profiles and abandonment timelines using decline curve analysis (DCA) for oil and pressure/z-factor methods for gas reservoirs [2].

A critical constraint in this study was the annual injection capacity ($k_{r,t}$), which defines the maximum allowable $CO_2$ injected into each reservoir per year. This parameter is derived from well performance simulations that reflect field-specific operational conditions and pressure constraints. This study focuses on eight oil and gas reservoirs with varying characteristics. Well simulations utilized fluid properties of gas and oil for each field from South Asia CCS opportunity mapping [55], while reservoir properties were based on the literature [41, 48, 49] for oil fields and [1, 8, 37, 43] for gas fields.

IPM-PROSPER was employed to simulate $CO_2$ injection under pressure-constrained conditions. The injection strategy followed the Porthos Project benchmark [5, 44], beginning with gas-phase injection and transitioning to liquid phase after reaching the critical pressure. Several technical assumptions were adopted in the simulation to reflect operational realities. Wellhead pressures were constrained to 50 bar for gas injection and 87 bar for liquid injection. Injection rates were further limited to ensure that reservoir pressure did not exceed 90% of the estimated fracture pressure, providing a conservative safety margin.

The integrated simulation for $CO_2$ injection was constructed using a compositional EoS PVT model to accurately capture phase behavior of the injected $CO_2$ and in-situ reservoir fluids across varying pressure and temperature conditions. For injectivity assessment, the Jones IPR (Inflow Performance Relationship) model was applied to evaluate near-wellbore pressure response and deliverability, incorporating non-Darcy flow effects and permeability-thickness dependencies to ensure realistic injection rate predictions under reservoir constraints. The simulation outputs include reservoir pressure buildup, well flowing pressure, and injection rate profiles. These results were analyzed to produce injectivity curves and pressure–storage efficiency relationships for each reservoir. The annual injection capacities ($k_{r,t}$), were calculated by aggregating allowable injection volumes over each simulation year, accounting for dynamic pressure behavior and operational constraints. The reservoir-specific injection limits over time were then implemented as upper bounds in the MILP optimization model. A representative result is shown for reservoir r3 in Fig. 4, while full outputs—including reservoir-specific injection limits,

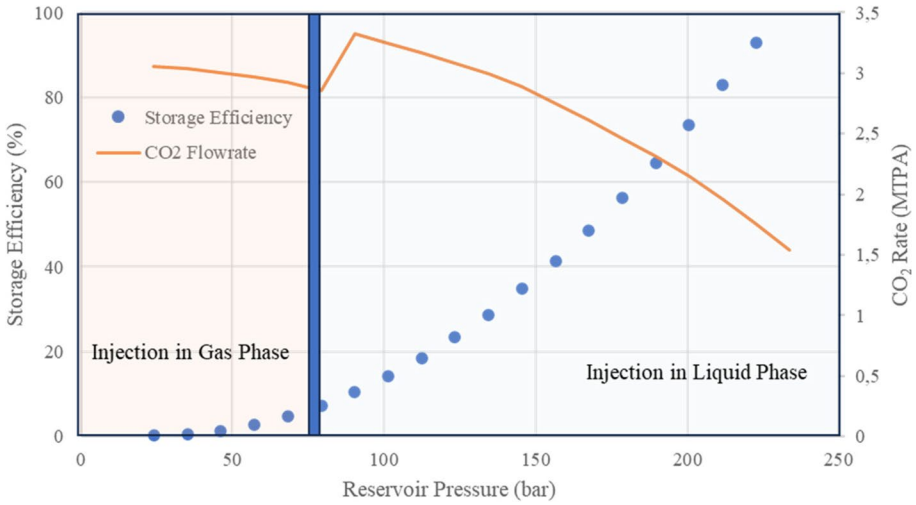


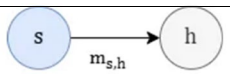
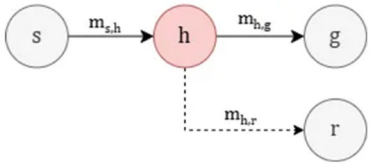
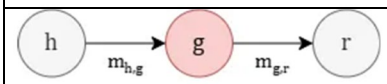
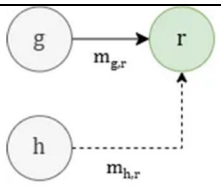

**Fig. 4** $CO_2$ injection rate vs storage efficiency for r3

storage costs, capacity, and availability—are provided in the Supplementary Materials (Table S2 and Fig. S1–S7).

Several key constraints were incorporated into the optimization model to ensure technical feasibility and operational reliability of the CCS network. These include annual reservoir injection capacity, reservoir readiness, and flow conservation at each node in the pipeline network. The mathematical representations of these constraints are summarized in Table 1.

Table 1 summarizes the mathematical constraints implemented in the CCS network model. The mass balance conditions ensure that all captured $CO_2$ ($m_{s,t}$) is transferred from sources through hubs and junctions to reservoirs, so that the mass stored in each reservoir ($m_{r,t}$) equals the total inflow. The annual injection capacity constraint requires that the injected mass ($m_{r,t}$) does not exceed the allowable limit ($k_{r,t}$) for each reservoir and year, while the readiness constraint

**Table 1** Optimization Constraints

| Component Representation | Descriptions | Mathematical Formulation |
|---|---|---|
|  | Mass distribution at sources | $\sum_{h\in \boldsymbol{H}} m_{s,h} = m_s, \quad \forall s \in \boldsymbol{S}$ |
|  | Mass balance at hubs | $\sum_{s\in \boldsymbol{S}} m_{s,h} = \sum_{g\in G} m_{h,g} + \sum_{r\in \boldsymbol{R}} m_{h,r}, \quad \forall h \in \boldsymbol{H}$ |
|  | Mass balance at junctions | $\sum_{h\in \boldsymbol{H}} m_{h,g} = \sum_{r\in \boldsymbol{R}} m_{g,r}, \quad \forall g \in \boldsymbol{G}$ |
|  | Mass stored at reservoirs | $\sum_{g\in \boldsymbol{G}} m_{g,r} + \sum_{h\in \boldsymbol{H}} m_{h,r} = m_r, \quad \forall r \in \boldsymbol{R}$ |
| Annual reservoir injection capacity | | $m_{r,t} \le k_{r,t}, \quad \forall r \in \boldsymbol{R}, \forall t \in \boldsymbol{T}$ |
| Reservoir readiness | | $m_{r,t} = 0, \quad \forall r \in \boldsymbol{R}, \forall t < Y_{r,ready}$ |

enforces $m_{r,t} = 0$ for all years prior to the reservoir's operational start year ($Y_{r,ready}$). Collectively, these constraints maintain mass conservation and operational feasibility throughout the network.

## 4 Results and discussion

The source–sink system for South Sumatra was optimized to determine the most cost-effective $CO_2$ storage network. To comprehensively evaluate alternative infrastructure and operational strategies, three distinct scenarios were developed, each incorporating different well configurations and injection scheduling approaches aimed at maximizing operational efficiency, cost-effectiveness, and system scalability. Table 2 summarizes the key features of each scenario. Detailed annual storage outcomes for each scenario are provided in the Supplementary Material (Fig. S8–S10). The following analysis presents results for each scenario sequentially, followed by a comparative discussion highlighting their key differences, scenario performances, and implications for large-scale CCS deployment.

### 4.1 Optimized networks injection scenarios

This section presents the optimized CCS network configurations generated under three alternative deployment scenarios: SW-A, MW-A, and MW-D. Each scenario is designed to address specific operational constraints, including limited early-stage reservoir availability, injection bottlenecks, and system-wide flow instability. The optimization period begins in 2027, which corresponds to the earliest year at which any reservoir becomes available for injection based on depletion estimation, and extends through 2056, covering a 30-year planning horizon. Scenario performance is evaluated in terms of injection coverage, network stability, and total system cost across the 30-year planning horizon.

Figures 5 and 6 illustrate the evolution of the SW-A network from 2027 to 2056. In the initial two years, as shown in Fig. 5a, the $CO_2$ storage network faced significant constraints, with only two reservoirs—r7 and included a direct connection to r8. The combined injection capacity of these reservoirs was only 2.05 MTPA, substantially below the target of 5.33 MTPA. This shortfall led to a considerable backlog, particularly for emitters with high $CO_2$ capture costs (greater than $90 per ton). These findings underscore the urgent need for improved infrastructure and capacity management to minimize delays and achieve emission reduction targets.Between 2027 and 2036, as depicted in Fig. 5b, the $CO_2$ storage network successfully expanded its capacity with the commissioning of additional reservoirs. The h2 emerged as the primary transport node during this period, driven by the exclusive availability of its connected reservoirs from 2027 to 2032. The direct connections of h1 to r4 and h2 to r8 eliminated intermediate junctions, reducing logistical complexities and improving the efficiency of the injection process. Starting in 2029, the network achieved full capture coverage, with 100% of captured $CO_2$ successfully injected into available reservoirs. The addition of new sinks such as r6, r3, r2, and r4, alongside the utilization of existing infrastructure, enabled a more balanced distribution of $CO_2$ across the network. The joint operation of h1 and h2 facilitated efficient routing and prevented any backlog, significantly enhancing the network's operational performance.

From 2027 to 2056, as illustrated in Fig. 6, the contributions of individual reservoirs shifted over time. Reservoir r3 quickly reached full utilization due to its small capacity, while r6 provided the largest storage contribution, supported by its high capacity and long operational period beginning in 2029. Although r5 played a minor role in the early years, it became a key injection site in the final 15 years, reflecting its high injection rate of 3.5 MTPA. In contrast, r3 was underutilized and received no further $CO_2$ after the first 10 years. By the end of 2056, h2 dominated $CO_2$ distribution, handling 72% of the total $CO_2$ flow.

Overall, these findings underscore the progressive optimization of the CCS network over time. Early operational constraints were gradually resolved, allowing a transition toward a more efficient and balanced distribution of storage contributions across the available reservoirs. The results highlight the importance of strategic planning in enhancing the resilience and cost-effectiveness of $CO_2$ storage systems. Nonetheless, the initial years of operation revealed significant challenges, including backlog caused by limited storage capacity.

In response to early-year injection constraints, the MW-A scenario was developed to enhance reservoir performance through targeted infrastructure expansion.

**Table 2** Summary of developed scenarios

| Scenario | Well configuration | Injection schedule |
|---|---|---|
| SW-A | Single well employed for all fields | h1 and h2 activated concurrently upon reservoir availability |
| MW-A | Multi-wells added to low-injectivity reservoirs | h1 and h2 activated concurrently upon reservoir availability |
| MW-D | Multi-wells added to low-injectivity reservoirs | h1 activation delayed to 2039; h2 follows initial schedule |

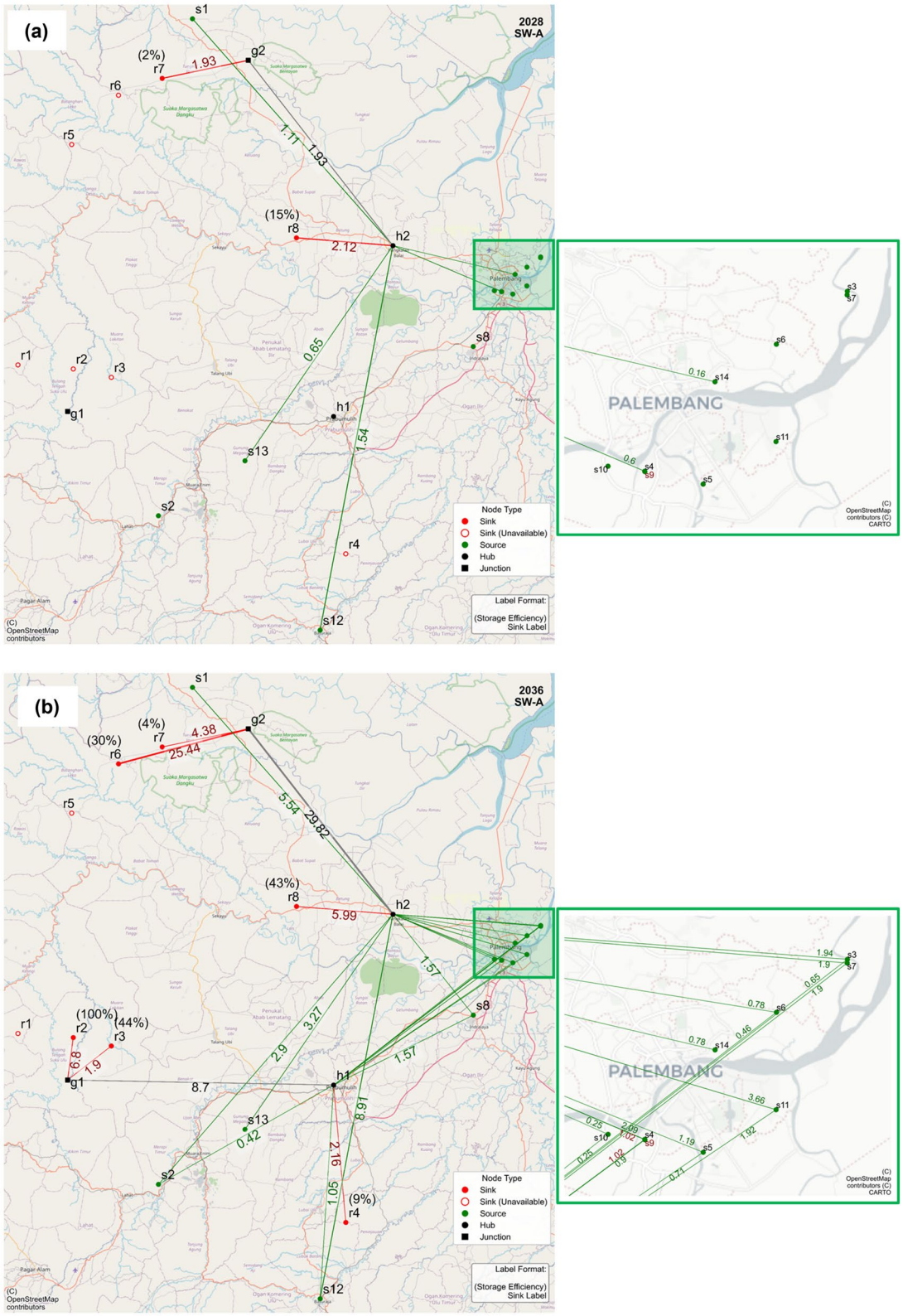



**Fig. 5** Optimized $CO_2$ network configuration for the SW-A scenario at (a) 2027–2028 and (b) 2027–2036, with lines representing selected transport pathways and numbers indicating transported $CO_2$ volume in MT $CO_2$

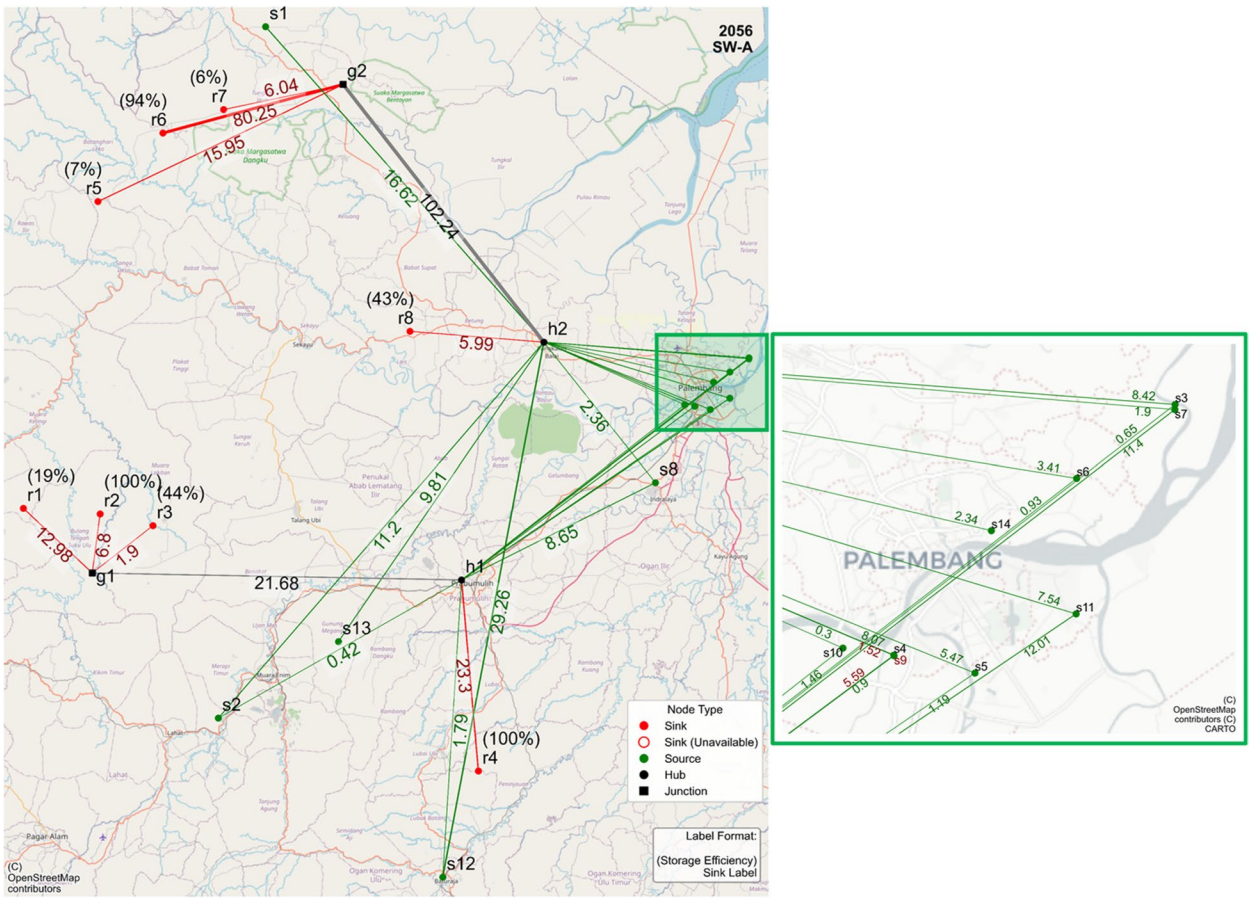



**Fig. 6** Optimized $CO_2$ network configuration for the SW-A scenario at 2027–2056, with lines representing selected transport pathways and numbers indicating transported $CO_2$ volume in MT $CO_2$

Specifically, four wells were added to reservoirs r1 and r7, and two wells to reservoir r8. Reservoirs r7 and r8 were selected due to their relatively low injectivity (below 1 MTPA), but they are among the earliest available for injection. Reservoir r1, while also exhibiting low injectivity, was prioritized for its substantial storage capacity of 68MT, positioning it as a critical asset for long-term system scalability. This configuration significantly increases early injection capacity, enabling the full capture of $CO_2$ emissions from the outset and mitigating delays and underutilization observed in the baseline scenario.

The transition from SW-A to MW-A demonstrates the substantial impact of network optimization and multi-well deployment on reservoir utilization, as shown in Fig. 7a and b. In MW-A, the implementation of multi-well configurations at r1 leads to a significant increase in storage capacity, enabling it to store almost three times more $CO_2$ compared to its role in SW-A. This underscores the effectiveness of multi-well strategies in unlocking greater injection potential from key reservoirs. Additionally, both r7 and r3 experience marked increases in their storage contributions, highlighting the benefits of enhanced injection infrastructure and improved network routing. Meanwhile, r4, which played a major role in SW-A, sees reduced utilization as capacity is reallocated to reservoirs with newly increased potential. Reservoir r6 and hub2 remain central in both scenarios, continuing to serve as primary distribution points for $CO_2$. While these results demonstrate that the adoption of multi-well strategies and targeted network enhancements in MW-A delivers substantial improvements in storage capacity, operational flexibility, and overall system performance compared to SW-A, they also reveal new operational challenges related to system stability.

The MW-A scenario improved system efficiency, resulting in a more balanced distribution of $CO_2$ storage between h1 and h2, with a ratio of 31:69. However, evaluation over the 30-year period, as shown in the stacked chart in Fig. 8, revealed periods of instability in $CO_2$ flow, particularly between 2033 and 2039. Although MW-A successfully met the $CO_2$ injection target and eliminated

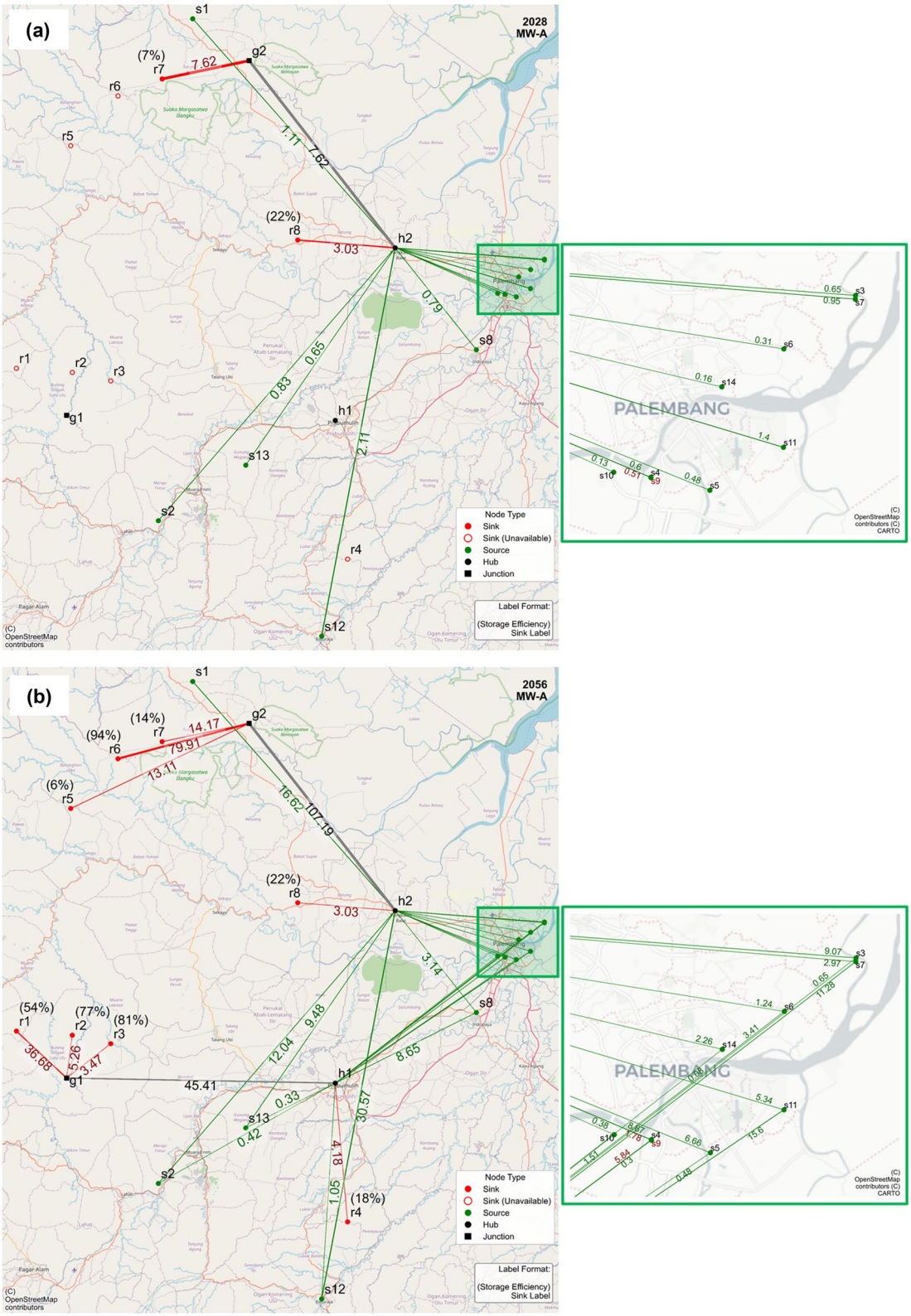



**Fig. 7** Optimized $CO_2$ network configuration for the MW-A scenario at (a) 2027–2028 and (b) 2027–2056, with lines representing selected transport pathways and numbers indicating transported $CO_2$ volume in MT $CO_2$

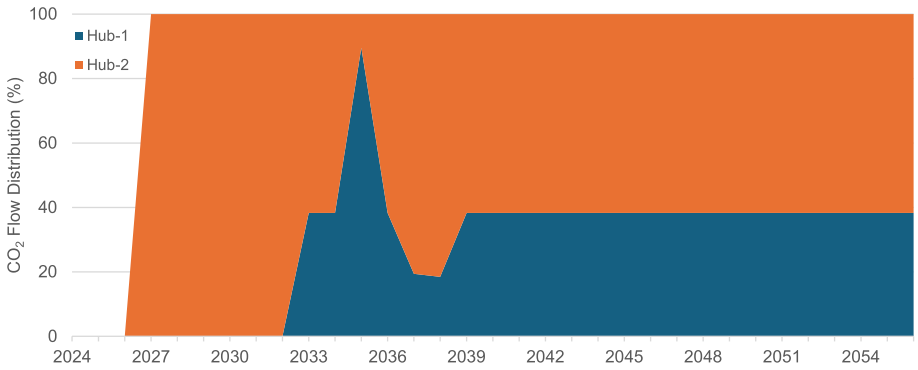



**Fig.8** $CO_2$ flow distribution in the MW-A scenario

backlog, these transitional instabilities highlight the need for further system coordination and operational optimization to ensure robust and reliable performance over time.

The mid-period fluctuations observed in MW-A motivated the development of the MW-D scenario, which aimed to improve system stability and better align reservoir deployment with infrastructure readiness. To address these issues, the MW-D scenario builds on the MW-A strategy by retaining the additional wells at reservoirs r1, r7, and r8. The key distinction of the MW-D scenario is the strategic delay in activating all reservoirs connected to h1, aligning their operation with the hub's full capacity, which becomes operational in 2039. This adjustment mitigates the fluctuations observed in the MW-A scenario by synchronizing the activation of reservoirs and hubs, ensuring a more stable $CO_2$ flow across the system. In contrast, h2 follows its original timeline, operating consistently and contributing to the overall system's stability. By optimizing activation timelines and expanding injection capacity, the MW-D scenario effectively addresses the operational challenges faced in MW-A, resulting in enhanced system efficiency (Fig. 9).

In the MW-D scenario, several key outcomes emerge, as illustrated in Fig. 10. Between 2027 and 2038, only h2 is operational, which leads to a reduction in the contributions from reservoirs r2, r3, and r4 compared to the MW-A scenario, despite their equivalent storage capacities. This reduction occurs because the activation of these reservoirs is intentionally delayed until 2039. By synchronizing reservoir activation with the operational capacity of h1, the MW-D scenario addresses the operational instabilities observed in previous configurations, particularly during the transitional period from 2033 to 2039. The strategic delay in activating reservoirs helps prevent storage imbalances and flow disruptions, resulting in a more balanced and efficient $CO_2$ storage system. As shown in the stacked chart in Fig. 11, this approach leads to a smoother and more consistent $CO_2$ flow over the 30-year period. The combination of enhanced injection capacity and optimized reservoir activation timelines positions the MW-D scenario as a more stable and efficient solution for $CO_2$ storage.

In summary, the three scenarios demonstrate distinct differences in infrastructure complexity, temporal alignment, and overall system performance. The SW-A configuration adopts a minimal design but experiences early injection backlogs due to limited reservoir availability. The MW-A scenario addresses this constraint by increasing early injection capacity through additional wells; however, it introduces mid-term flow variability as a result of asynchronous infrastructure deployment. The MW-D scenario mitigates this issue by aligning reservoir activation with hub readiness, resulting in more stable injection dynamics throughout the planning horizon. While MW-A improves early-phase storage performance, MW-D achieves greater consistency and efficiency in network utilization over time. These findings provide valuable guidance for optimizing CCS network planning and infrastructure deployment strategies in large-scale industrial clusters.

### 4.2 CCS cost

Optimization results across the three scenarios (SW-A, MW-A, and MW-D) revealed a consistent pattern in the total costs and cost distribution, despite differing strategies for $CO_2$ injection and storage. The total costs for the scenarios range from $17.53 billion to $18.15 billion over 30 years, whether a single well is used (SW-A) or multiple wells are deployed (MW-A and MW-D). This suggests that adjustments to injection strategies, such as increasing the number of wells, do not significantly increase overall expenditure. Instead, these strategies optimize logistical aspects, such as enhancing storage capacity and

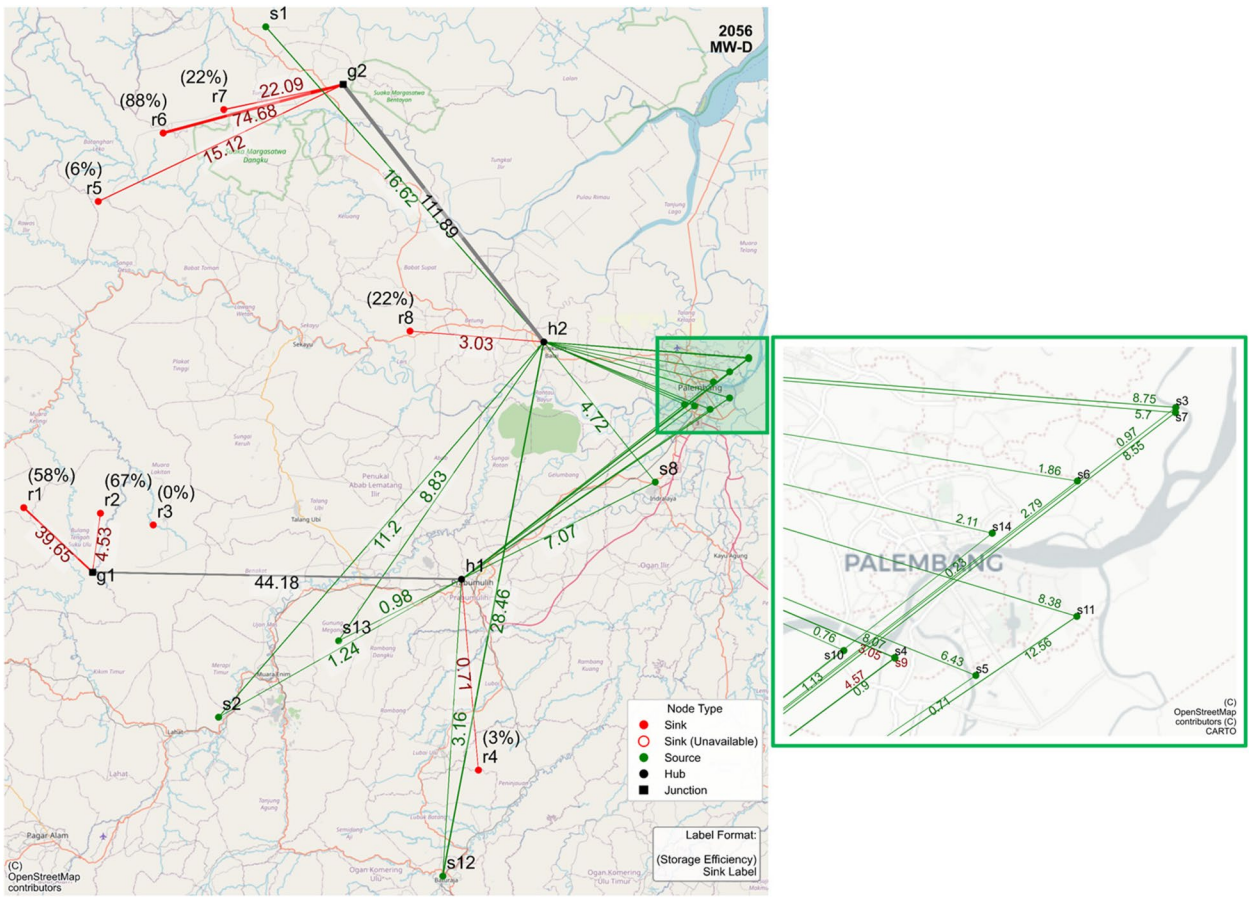



**Fig. 9** Optimized $CO_2$ network configuration for the MW-D scenario at 2027–2056, with lines representing selected transport pathways and numbers indicating transported $CO_2$ volume in MT $CO_2$

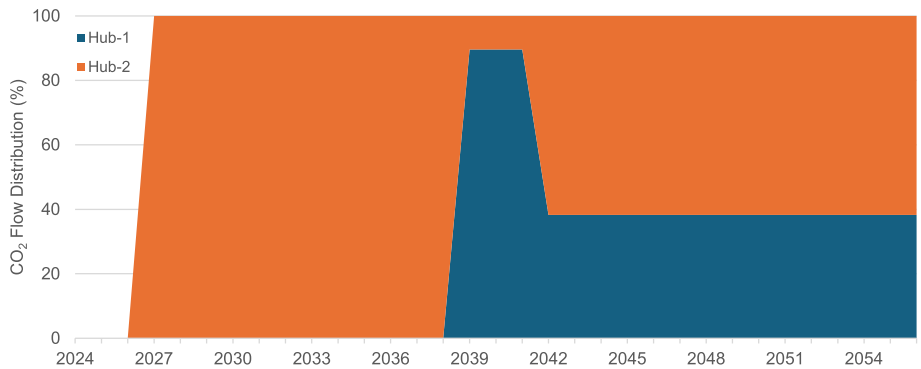



**Fig. 10** $CO_2$ flow distribution in the MW-D scenario

reducing transportation costs. In the MW-A and MW-D scenarios, the use of multi-well configurations provides significant operational advantages, despite the initial higher deployment costs. While the addition of wells increases infrastructure costs, these are offset by the reduction in backlog during the early years of operation. For example, between 2027 and 2028, when only two reservoirs are available for injection, the addition of wells in reservoirs r7 and r8 allows the system to meet injection targets efficiently. This strategy reduces reliance on limited capacity and prevents delays in $CO_2$ storage,

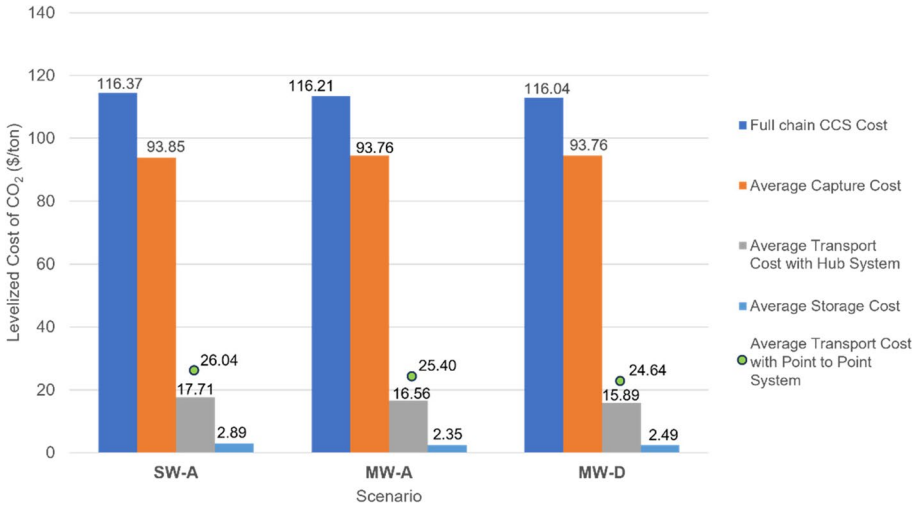



**Fig. 11** Comparison of CCS costs breakdown by different scenarios

ultimately avoiding additional costs and improving overall system performance.

Figure 11 presents a comparative analysis of the three scenarios over a 30-year period. In the SW-A scenario, a single well was used across all fields, resulting in the storage of 153.21 MT of $CO_2$ at a total cost of $17.53 billion. In contrast, the MW-A and MW-D scenarios introduced multi-wells, increasing total $CO_2$ stored to 159.81 MT, while total costs remained around $18 billion. The reduced storage capacity in SW-A (153.21 MT) was primarily due to limitations during the early years of operation, which restricted the system's ability to meet injection targets. In all scenarios, capture cost are the dominant component, accounting around 80% of the total full chain CCS Cost.

Transportation costs per ton were consistently lower in the multi-well scenarios compared to SW-A, decreasing from $17.71 per ton in SW-A to $16.56 and $15.89 per ton in MW-A and MW-D, respectively. These reductions reflect the impact of more efficient network configurations—such as the deployment of additional wells and optimized reservoir activation—which improve logistical flexibility and enhance the utilization of shared infrastructure.

Analysis of $CO_2$ transportation costs further highlights that hub-based systems are significantly more efficient than point-to-point systems. The average transportation cost decreases from approximately $25.36/ton in point-to-point systems to $16.72/ton in hub-based systems, primarily as a result of economies of scale and shared pipeline networks. By enabling multiple $CO_2$ sources to access common infrastructure and carefully managing injection capacity and timing, hub systems minimize the need for separate pipelines and improve overall cost efficiency.

The integration of MILP with detailed reservoir availability times and injection rate calculations from well simulations refines the $CO_2$ injection and storage process. This method enables better synchronization between reservoir activation and injection capacity, providing a more precise framework for optimizing storage strategies. Ultimately, this approach enhances both system efficiency and cost-effectiveness, contributing to the scalability and reliability of $CO_2$ storage systems in large-scale implementations.

### 4.3 Cost sensitivity analysis of CCS network configuration

This section evaluates the robustness of the optimized CCS network configuration under cost sensitivity, using the multi-well deployment (MWD) scenario with full storage enforcement. In this setting, all captured $CO_2$ must be permanently stored, reflecting a policy context where venting or partial capture is not permitted. The optimization process is therefore constrained to route fixed volumes of $CO_2$ through the network while minimizing total system cost.

Capture capital costs were increased by 30% to reflect typical uncertainty in early-stage project development, following the factored estimate approach based on major equipment costs and with a probable accuracy of up to ± 30% [26, 38]). As summarized in Table 3, this adjustment resulted in a higher full chain CCS cost, but did not affect the main sinks utilized or the overall network configuration. Notably, because capture costs contribute over 75% of the total full chain cost, an increase in capture CAPEX leads to a significant rise in total system cost—up to 13.6%. Since capture is modeled as a fixed requirement, changes in its cost do not influence the structural arrangement of the network.

**Table 3** Impact of cost sensitivity on CCS network configuration

| Sensitivity Scenario | Full Chain Cost Change (%) | Main Sinks Utilized | Notes |
|---|---|---|---|
| MW-D + 30% Capture CAPEX | + 13.6 | r6, r1, r7 | No change in network configuration |
| MW-D + 30% Transport Cost | + 4.2 | r6, r1, r7 | No change in network configuration |
| MW-D + 30% Uniform Storage Cost | + 0.7 | r6, r1, r7 | No change in network configuration |
| MW-D + 50% Selective Storage Cost (r1, r6) | + 0.6 | r5, r1, r6 | Allocation shifted |

Transport cost sensitivity was evaluated by increasing pipeline and booster infrastructure costs by 30%, reflecting uncertainties in terrain, material prices, and construction complexity. Although this scenario led to higher system costs, the main sinks and network configuration remained unchanged. Since transport accounts for less than 15% of total system costs, the resulting overall cost increase was modest at 4.2%.

Storage cost sensitivity was assessed through both uniform and selective adjustments. Uniform increases in storage costs, which account for less than 10% of total system cost, resulted in only a slight increase in the full chain cost and did not impact on the network structure or the allocation of main sinks. However, when a selective cost increase was applied to specific reservoirs (r1 and r6), the model responded by reallocating $CO_2$ volumes to alternative sinks, notably r5. These findings indicate that, while storage costs are a relatively small portion of the total cost, spatial cost variations can prompt allocation shifts within the network if they are substantial enough to shift the economic competitiveness of storage options.

Overall, the results demonstrate that the network structure under the MW-D storage scenario is primarily governed by spatial constraints, sink availability, and time-dependent injection capacity. While total expenditure increases with rising costs, the structural arrangement of the network remains stable unless significant cost escalations shift the economic competitiveness of individual storage sites.

### 4.4 Implementation considerations and broader applicability

While this study focuses on cost optimization, broader CCS implementation also depends on environmental risk management, public engagement, and effective regulation. In Indonesia, requirements for $CO_2$ leakage monitoring and storage safety are increasingly addressed through dedicated policies such as MEMR Regulation No. 2/2023. The Gundih project, as Indonesia CCS pilot initiative, has been specifically studied for its public acceptance aspects. Evidence shows that public acceptance is more likely when developers prioritize transparent communication, active stakeholder engagement, and continuous efforts to understand local perceptions. Understanding public perceptions and implementing effective engagement and communication strategies with local communities are crucial for the successful implementation of CCS projects [33]. In South Sumatra, where oil and gas industries form the backbone of the local economy, CCS provides a pathway to maintain and even extend the operational lifespan of existing oil and gas activities. Rather than causing disruption, integrating CCS enables these activities to continue in line with climate targets, helping to preserve local employment and economic stability as the region transitions to a lower-carbon future.

Beyond this regional context, the modeling framework developed in this study is broadly applicable to other CCS planning scenarios, particularly for onshore systems with a spatial extent of less than 300 km radius and utilizing pipeline-based transport. Its flexible, superstructure-based MILP formulation can be adapted to various geographic settings, emission patterns, and infrastructure constraints. The model is capable of handling spatial clustering of sources, phased storage availability, and staged investment, which are common in both industrial clusters and national CCS programs. By integrating spatial and temporal dimensions, the framework supports scenario analysis, investment prioritization, and system optimization under dynamic conditions. This versatility makes it a valuable decision-support tool for stakeholders seeking to develop resilient and cost-effective CCS networks that can be tailored to local needs and evolving policy environments.

### 4.5 Model limitations

The technical and economic parameters in this study are derived from published literature and process simulation results, providing a reasonable basis for system-scale CCS modeling. The analysis applies a fixed interest rate of 8%, which aligns with values commonly used in developing economies, while acknowledging that future market conditions and risk profiles may change. Reservoir decline curves and abandonment pressures were determined using established industry references. Reservoirs are represented as homogeneous, with the understanding that

site-specific heterogeneity and structural features could affect actual storage performance and injection rates. Pipeline network design utilizes currently available infrastructure data, without explicit modeling of potential future expansions, topographical variations, or regulatory factors that may influence costs and routing. These modeling choices reflect standard practices for system-level assessments, supporting comparative scenario analysis and optimization. As more detailed and site-specific data become available, this framework can be refined to further enhance its relevance for CCS planning and policy development.

## 5 Conclusions

This study optimizes CCS systems in Sumatra using a multiperiod MILP model, incorporating detailed well simulations and abandonment prediction calculations. These innovations enabled the precise determination of injection start times and annual capacities, which are essential for optimizing hub configurations and maximizing $CO_2$ storage efficiency. The SW-A scenario, utilizing a single-well configuration without activation delays, encountered challenges such as $CO_2$ injection backlogs and operational instability, particularly during the early years. Despite achieving a storage capacity of 153.21 MT at a total cost of $17.53 billion, the system's efficiency was limited by suboptimal storage utilization and delays. In contrast, the MW-A scenario, which introduced additional wells, and the MW-D scenario, which combined additional wells with strategically delayed reservoir activations, successfully addressed these challenges. These changes increased $CO_2$ storage to 159.81 MT, with only a marginal cost increase to $18.03–18.12 billion. Notably, transportation costs per ton were reduced from average $25.36 in point-to-point systems to $15.89–17.71 in hub systems, resulting in a 34% cost reduction attributable to economies of scale and shared infrastructure. This analysis demonstrates the novelty of integrating MILP with more detailed and realistic constraints, specifically the injection starts times and annual capacity calculations that accurately represent reservoir characteristics. By incorporating these refined constraints, the model captures the dynamic availability and economic viability of each reservoir over time, providing a more robust framework for optimizing $CO_2$ storage allocation. This approach enables a deeper understanding of how reservoir-specific factors, such as injection timing and capacity evolution, impact overall network performance, offering valuable insights for designing efficient and practical CCS systems.

**Nomenclature**

| | |
|---|---|
| $S$ | Set of sources $(s = \{1,2,\ldots,14)$ |
| $H$ | Set of hubs $(h = \{1,2\})$ |
| $G$ | Set of junctions $(g = \{1,2\})$ |
| $R$ | Set of reservoirs $(r = \{1,2,\ldots,8)$ |
| $T$ | Set of years in the 30-year period $(t = \{1,2,\ldots,30\})$ |
| $m_{s,t}$ | Mass flow captured $(\mathrm{MT})$ in source $s$ in year $t$ |
| $m_{s,h,t}$ | Mass flow rate $(\mathrm{MT})$ from source $s$ to hub $h$ in year $t$ |
| $m_{h,g,t}$ | Mass flow rate $(\mathrm{MT})$ from hub $h$ to junction $g$ in year $t$ |
| $m_{g,r,t}$ | Mass flow rate $(\mathrm{MT})$ from junction $g$ to reservoir $r$ in year $t$ |
| $m_{h,r,t}$ | Mass flow rate $(\mathrm{MT})$ from hub $h$ to reservoir $r$ in year $t$ |
| $m_{r,t}$ | Mass flow stored $(\mathrm{MT})$ in reservoir $r$ in year $t$ |
| $C_s^{\text{capture}}$ | Capture cost per mass $(\$/\mathrm{ton})$ captured at source $s$ |
| $C_{s,h}^{\text{transport}}$ | Cost $(\$)$ for transport from source $s$ to hub $h$ |
| $C_{h,g}^{\text{transport}}$ | Cost $(\$)$ for transport from hub $h$ to junction $g$ |
| $C_{g,r}^{\text{transport}}$ | Cost $(\$)$ for transport from junction $g$ to reservoir $r$ |
| $C_{h,r}^{\text{transport}}$ | Cost $(\$)$ for transport from hub $h$ to reservoir $r$ |
| $C_r^{\text{storage}}$ | Storage cost per mass $(\$/\mathrm{ton})$ injected to reservoir $r$ |
| $k_{r,t}$ | Annual injection capacity (MT) of reservoir $r$ in year $t$ |
| $Y_{r,ready}$ | The ready time of reservoir $r$, representing the earlies year when storage is allowed |

## Supplementary Information

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Supplementary Material 1.

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**Data availability**
The data and material analyzed in this current study are available from the corresponding author on reasonable request.

## Declarations

**Ethics approval and consent to participate**
Not applicable.

**Competing interests**
The authors declare no competing financial interests.

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